

Auditable Flourishing v6.2 External Validation, Burden, and Decision-Value Protocol

Status and purpose

This document governs empirical claims about Auditable Flourishing. It is designed to permit null, adverse, burden, capture, architecture-bias, and retirement findings. Internal verification, synthetic examples, and author-administered exercises cannot establish reliability, validity, fairness, legitimacy, decision value, or superiority.

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1. Governing questions

The programme asks:

1. Can trained reviewers apply each criterion, dimension, and final decision layer with adequate agreement and interpretable disagreement?
2. Does AF detect substantive protection rather than documentation polish?

3. Does it improve decisions, defect localization, uncertainty disclosure, rights routing, or justified refusal relative to simpler alternatives?
4. Does it create false clearances, false rejections, delay, burden, access failure, or architecture-linked error?
5. Can independent hosts reproduce administration and analysis without author or sponsor control?
6. Do affected parties find the process usable, fair, and capable of correction?
7. Which components should be retained, revised, restricted, paused, or retired?

2. Independence prerequisites

Before unblinding, publish or register:

- protocol, candidate, scenario, codebook, and analysis versions;
- sponsor, host, data, reviewer, adjudication, analysis, and publication control;
- host interests and preferred outcomes;
- reviewer-pool construction and selection;
- immutable rating-deposit system;
- conflicts, recusals, and AI assistance;
- candidate inclusion and non-participation;
- affected-party selection, compensation, access, and withdrawal;
- null, adverse, and protocol-failure publication rights;
- amendments, stopping, and security redaction.

An author or sponsor veto invalidates an independent-validation claim.

2A. Study readiness tiers

Tier	Minimum evidence	Permitted statement
Design-ready	protocol, templates, draft scenario plan, training/analysis/burden/publication specifications	materials can be constructed and challenged
Materials-ready	frozen library, hashes, practice bank, qualification rubric, deposit system, scripts, capacity plan	dry rehearsal
Shadow-ready	host/reviewers contracted, conflicts/custody/deposits/appeal/publication rehearsed	non-dispositive study
Pilot-ready	live-scope legal, affected-party, adverse-event, remedy, stopping-rule conditions	bounded pilot under separate authority
Replication-ready	frozen post-pilot package and second-host transport plan	external replication

AF v6.2 is design-ready, not materials-ready.

3. Phased programme

Phase 0 - material construction, feasibility, and exploratory calibration

One independent host uses 20 frozen synthetic cases, four to six trained reviewers, the Stage A criteria, selected Stage B relation cases, the audit-theatre arm, immutable deposits, criterion-order randomization, global-impression probes, and full burden logging. It is exploratory. It estimates prevalence, completion, disagreement, interval width, order effects, missingness, scenario performance, false-green instrument behavior, and operational cost.

Phase 0 may be sequenced without changing the AF specification. Phase 0A tests Stage A criterion interpretation, final Stage A routing, independence administration, deposit integrity, audit-theatre discrimination, access, burden, and protocol-failure handling. Phase 0B introduces full Stage B anchor calibration and perturbation replay only after Phase 0A shows that the Stage A and administration layers are usable enough to support the added comparison burden. This sequencing does not remove Stage B or permit a Stage A-only superiority claim.

Phase 0 does not include live affected parties, cannot establish architecture fairness or domain validity, and does not justify a confirmatory threshold. Its result is retain, revise, restrict, pause, or retire the next phase.

Phase 0 precision boundary

At an illustrative agreement proportion of 0.80, twenty independent cases yield an approximate 95 percent interval of 0.62 to 0.98 before clustering and prevalence adjustments. Adding reviewers expands disagreement information but does not create independent cases. Final Stage A outcomes are reported descriptively at this scale unless the preregistered method establishes adequate support for a chance-adjusted coefficient. The independent host may increase the frozen case set to 30–40 if resources permit, without weakening any independence, deposit, scenario, access, or publication requirement.

Phase 1 - independent multi-arm pilot

At least one unaffiliated candidate and three materially different architecture families are included. Strong comparator arms, affected-party participation, multiple domains or contexts, and independent replay are introduced. Sample size is selected by simulation or decision-error requirements informed by Phase 0.

Phase 2 - external replication and shadow-mode decision study

At least two unaffiliated hosts replay the frozen protocol and scenario library. Where lawful and ethical, cases are shadowed against real decisions without determining entitlements or rights. Downstream outcomes, remedy, burden, and institutional capture are measured.

Phase 3 - bounded live evaluation

Live consequence-bearing use requires separate legal, ethical, institutional, community, safety, and deployment authorization. AF itself cannot grant this transition.

4. Comparator arms

The minimum Phase 1 design includes:

- AF v6.2;
- a simpler structured checklist;
- domain standard of care;
- a minimal-documentation control;
- an audit-theatre decoy with polished evidence but deliberately varied substantive safeguards;
- a preregistered no-AF or ordinary-process comparator where ethical and lawful.

Comparator fidelity, reviewer training, exposure time, and information access are recorded. A weak straw comparator cannot support superiority.

5. Scenario and candidate diversity

Scenario sets are controlled by the Scenario Governance and Calibration Standard. Candidates and cases vary architecture, domain, language, documentation intensity, resource level, disability and access needs, institutional form, community governance, reversibility, rights exposure, and uncertainty. Candidate non-participation and reasons are reported; the sample cannot be described as representative if only highly prepared entrants volunteer.

6. Primary outcomes

Outcomes are reported by criterion, dimension, and decision layer:

- exact and chance-adjusted agreement with uncertainty intervals;
- independent replay agreement;
- indeterminate, refusal, non-admissibility, admissibility, dominance, incomparability, and non-decisive rates;
- base relation versus perturbation-invariant outcome;
- false clearances and false rejections against a preregistered adjudicated reference where one is defensible;
- defects discovered, patch quality, appeals, reversals, and remedy;
- uncertainty reduction and decision changes;
- time, direct cost, uncompensated burden, access failure, withdrawal, and abandonment;
- affected-party value and procedural fairness;
- host interference, sponsor interference, publication pressure, and capture events;
- architecture-, language-, documentation-, disability-, and resource-linked differential performance;
- downstream outcomes and unanticipated harms where the phase permits.

No single coefficient or composite score can establish validation.

7. Decision value

Decision value compares the baseline decision with the AF-informed decision and records whether AF:

- changes the decision;
- prevents a false clearance;
- creates a false rejection;
- identifies a material defect;
- improves uncertainty or rights disclosure;
- causes delay or opportunity cost;
- improves remedy or appeal;
- creates institutional or affected-party burden;
- changes downstream outcomes.

Positive, neutral, and adverse examples are mandatory. A protocol that produces more documentation without changing decision quality or transparency may have no net value.

8. Reliability and precision

Phase 0 estimates, but does not confirm, agreement. Criterion-level ordinal ratings use a preregistered ordinal coefficient such as Gwet's AC2 when justified; nominal final outcomes use an appropriate nominal coefficient. The final conjunction is analyzed directly and is not inferred by multiplying criterion coefficients.

The confirmatory phase uses simulation based on observed prevalence, missingness, reviewer number, and correlation to select cases sufficient for the preregistered precision or decision-error objective. Thresholds, interval methods, multiplicity, subgroup analyses, and consequences of misses are fixed before data are unsealed.

9. Halo, order, and AI-assistance analyses

Criterion order is balanced or randomized. Global impressions are recorded before and after criterion rating. The analysis estimates whether order or broad impression explains agreement or adverse findings.

AI use is separated into retrieval-only, clerical, interpretive, rating-suggestive, and rating-generative conditions. Ratings materially influenced by a shared model are not counted as independent human ratings. Model, version, instructions, evidence exposure, and verification are recorded. AI-assisted and human-only results are compared separately.

10. Audit-theatre experiment

The audit-theatre arm holds presentation quality approximately constant while varying substantive safeguards, or varies polish while holding substantive protection constant. AF fails this experiment if polished but causally empty documentation receives materially better outcomes, lower uncertainty, or less burden than functionally protective but less polished evidence.

11. Architecture fairness and invariance

The programme tests whether criteria and dimensions have comparable meaning, missingness, burden, and error across architecture families and groups. Equal form administration is not sufficient. Differential item functioning, qualitative cognitive interviewing, error decomposition, architecture-specific non-applicability, and affected-party interpretation may be used. Failure can require assistance redesign, scope restriction, anchor revision, or criterion retirement.

The genealogy test is symmetric: both unaffiliated exclusion and affiliated advantage are assessed.

12. Ablation strategy

Exhaustive one-at-a-time removal of all criteria, dimensions, records, and procedures is not feasible as an initial study. The programme uses a staged ladder:

1. safety-critical ablations: C1, C5, burden gate, independence, immutable deposits, and perturbation replay;
2. audit-theatre and documentation-neutrality ablations;
3. prespecified grouped ablations aligned with the four Stage B functional groups;
4. fractional factorial, D-optimal, or adaptive designs for remaining components;
5. confirmatory targeted ablations only for components whose incremental value remains uncertain.

A component can be retained because it prevents rare severe error even when average decision value is small; the justification must be explicit and non-arithmetic.

13. Cost and feasibility

The cost model separates fixed and variable resources:

- governance, ethics, preregistration, scenario creation, training, infrastructure, and analysis setup;
- reviewer, adjudicator, candidate, affected-party, translator, accessibility, data, security, and publication hours;
- direct payments, software or hosting, travel, interpretation, legal review, and contingency;
- uncompensated time and opportunity cost.

Capacity is conjunctive. A design is not feasible merely because average hours per case are acceptable. The Phase 0 protocol provides the executable minimum and a formula-based budget template. If required access support, reviewer count, calendar, data control, or publication independence cannot be funded, the study narrows, pauses, or refuses.

Non-normative planning illustration. A prospective host can estimate rating workload as cases x independent reviewers x expected rating hours per case, then add training, calibration, adjudication, translation/accessibility, analysis, appeal, and publication separately. Twenty cases x four reviewers x 1.0-1.5 hours implies roughly 80-120 reviewer-hours before those additional functions. The example is not an AF requirement or an empirical cost claim; every assumption and range must be declared and replaced by observed burden data.

14. Adverse events and stopping

Immediate review or stopping follows material rights risk, retaliation, coercion, data breach, protected-evidence exposure, sponsor or host manipulation, publication veto, rating-deposit compromise, scenario leakage, high false-clearance rate, architecture-linked harm, excessive burden, or inability to distinguish candidate and protocol defects.

Stopping rules, who can trigger them, appeal, participant support, notification, and publication of the stopping reason are preregistered.

14A. Perturbation and factor-structure diagnostics

Record tie rate; proposed, admitted, rejected and reversal-capable perturbation rates; base/final relation distributions; and the share of base dominance surviving replay. Any tie/perturbation forecast is model-dependent and reproduced from versioned simulation code. Where sample size permits, compare one-factor and correlated multidimensional representations; C3-D07 applies when a simpler structure preserves floors and defect localization while improving reliability or burden, or dimension-specific residuals add no reproducible information.

15. Yield hypotheses

The study preregisters competing hypotheses rather than selecting one favorable prediction:

- Stage A range restriction raises base-profile dominance;
- defended alternatives convert some base dominance to non-decisive;
- stable crossing remains incomparable;
- low dominance yield can coexist with positive decision value;
- high non-decisiveness can reflect genuine uncertainty or defective anchors and must be diagnosed, not celebrated.

The analysis publishes base and invariant relations, reason codes, and sensitivity to scenario and anchor assumptions.

16. Registered Report route

A Stage 1 Registered Report is a preferred, not mandatory, route for Phase 0 or Phase 1 because methods, hypotheses, analysis, and publication commitment can be reviewed before results are known. The release

includes a submission-ready Stage 1 outline. No journal acceptance is claimed. If a Registered Report is not used, equivalent preregistration, independent publication control, and null/adverse-result rights remain mandatory.

17. Decision rule

At each phase, an independent body issues one of:

- **retain;**
- **revise;**
- **restrict;**
- **pause;**
- **retire;**
- **indeterminate.**

The decision cites reliability, validity, fairness, burden, access, decision value, adverse events, capture, and affected-party findings. Positive average performance cannot compensate for a material rights or catastrophic-risk failure.

Cited works

Center for Open Science. (n.d.). Registered Reports. <https://www.cos.io/initiatives/registered-reports>

Gwet, K. L. (2014). *Handbook of inter-rater reliability* (4th ed.). Advanced Analytics.

Royal Society Open Science. (n.d.). Registered Reports. <https://royalsocietypublishing.org/rsos/registered-reports>

AF v6.2 synchronization note

This artifact implements the v6.2 three-axis record (decision_status, evidence_state, review_state), the non-erasure invariant, perturbation-replay anti-inflation and anti-suppression controls, and the separation of raw agreement, reliability, adjudication, and external validity. These controls clarify the existing protocol and do not confer legal, moral, procurement, or deployment authority.